

HW 4: Camera THU

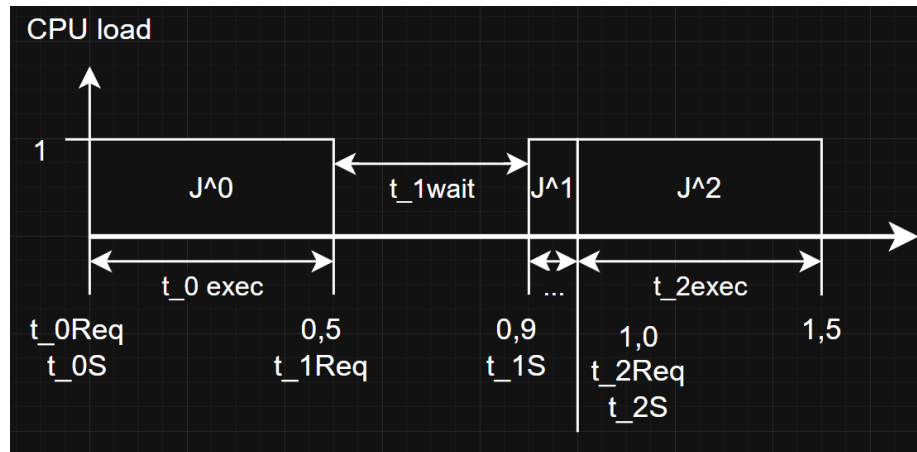
a) Jobs

Numbers are given in seconds

Camera initialization: $J^0 = (0, 0.9)$

Expose photo: $J^1 = (0.9, 0.1)$

Store photo: $J^2 = (1, 0.5)$



The “...” represents “t₁ exec”

b)

b1)

Camera initialization: $J^0 = (0, 0.9)$

Expose and Store photo: $J^1 = (0.9, 0, 6)$

b2) & b3)

$$t_{rate_min} = 1 \div (t_{exec} + t_{wait}) = 1 \div 1,5s \approx \underline{0,67s^{-1}}$$

$$t_{rate_max} = 1 \div t_{exec} = 1 \div 1s = \underline{1s^{-1}}$$

b4) & b5)

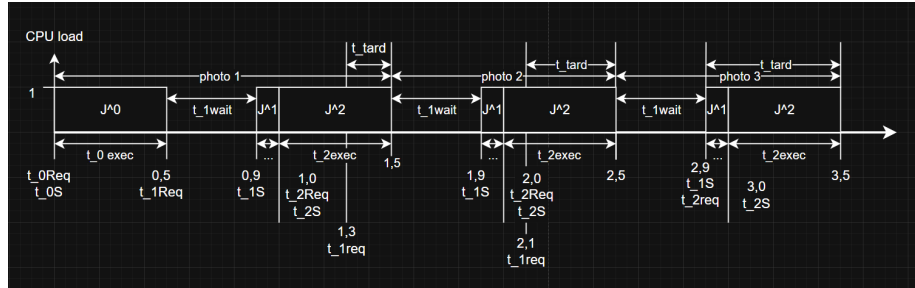
$$t_{per} = 0,8s$$

$$t_{exec} = 1s$$

$$t_{exec} > t_{per} \Rightarrow \text{tardiness over } 0$$

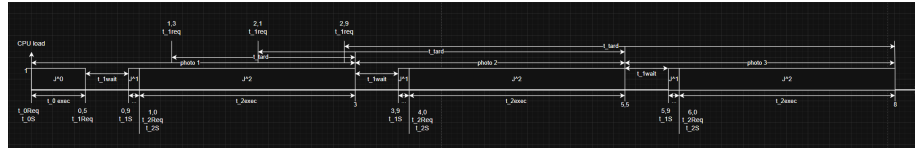
Tardiness per pictures taken x :

$$t_{tard}(x) = (t_{exec} - t_{per})x$$



All dead- & completion times are implied to be at the end of each execution time

c)



$$t_{rate_min} = 1 \div (t_{exec} + t_{wait}) = 1 \div 3s \approx 0,33s^{-1}$$

$$t_{rate_max} = 1 \div t_{exec} = 1 \div 2,5s = 0,4s^{-1}$$